

Probability Theory

Though there be no such thing as chance in the world, our ignorance of the real cause of any event has the same influence on the understanding.

—DAVID HUME

Our book begins by providing a formalization of probability theory, establishing some basic theorems that will allow us to formally describe random generative processes and quantify relevant features of these processes. We believe that it is important for researchers to understand the assumptions embedded in mathematical probability theory before attempting to make statistical claims. In keeping with the agnostic paradigm, we will attempt to draw attention to the relevant intricacies involved, and highlight where mathematical constructions serve as approximations. The foundation of probability theory allows researchers to precisely define the scope of their research questions and to rigorously quantify their uncertainty about the conclusions they draw from their findings. Our approach is somewhat unconventional, in that we focus on describing random variables *before* we consider data, so as to have well-defined inferential targets. This is sometimes referred to as the “population first” approach.¹ This approach will enable us to engage with the more complex topics in Parts II and III with relative ease and rigor.

We begin this chapter with a discussion of how random generative processes assign probabilities to random events. We can then describe probabilities of individual events and how the probability of one event relates to the probability of another. We proceed to consider random variables, which take on real number values determined by the outcomes of random generative processes. We describe several types of functions that characterize the probability *distributions* of random variables; these distribution functions allow us to characterize the probability that the random variable takes on any given value

¹ See Angrist and Pischke (2009).

or values. When we have two or more random variables whose values are determined simultaneously, we can describe their joint distributions, which also allows us to describe their marginal and conditional distributions. Here, we primarily focus on bivariate relationships (between two random variables), but we outline multivariate generalizations. At the end of the chapter, we provide suggestions for readers interested in a more technical treatment of the material or alternative approaches.

1.1 RANDOM EVENTS

Probability theory is a *mathematical construct* used to represent processes involving randomness, unpredictability, or intrinsic uncertainty. In a setting in which there are several possible outcomes, each with some probability of occurring, we refer to the process by which the outcome is determined as a *random generative process*. In this section, we present the basic principles of probability theory used to describe random generative processes.

1.1.1 What Is Probability?

We can think of a random generative process as a mechanism that selects an *outcome* from among multiple possible outcomes. This mechanism could be flipping a coin or rolling a die, drawing a ball from an urn, selecting a person at random from a group of people, or any other process in which the outcome is in some sense uncertain. A single instance of selecting an outcome is known as a *draw* from or *realization* of the random generative process. The term *experiment* is also commonly used, but we shall refrain from this usage to avoid confusion with experiments in the ordinary sense of the term.

The probability of an event describes the proportion of times that event can be expected to occur among many realizations of a random generative process. This interpretation of probability is known as *frequentist probability* or *frequentism*: the probability of an event A is interpreted as representing how frequently A would occur among many, many draws from a random generative process. It is the long-run average or limiting value of the frequency of observing event A among repeated realizations of the generative process.²

Probability theory is a *model*, which is an approximation of reality. Everyday macrophysical processes are not actually characterized by fundamental randomness. Consider, for example, a coin flip. In principle, if we could know the exact mass, shape, and position of the coin at the moment it was flipped and the exact magnitude and direction of the force imparted to it by the flipper

² There are other interpretations of probability, most notably the *Bayesian* interpretation, which treats probability as representing a degree of belief or confidence in a proposition. We do not discuss these alternative interpretations and will be operating under the frequentist paradigm throughout this book.

and of all other forces acting on it, we could predict *with certainty* whether it would land on heads or tails.³

The mathematical construct of randomness is, therefore, a *modeling assumption*, not necessarily a fundamental feature of reality.⁴ It allows us to model the outcomes of the coin flip given our uncertainty about the exact nature of the forces that will act on the coin in any particular instance. Similarly, in the social and health sciences, the assumption of randomness allows us to model various outcomes that we might care about, given our uncertainty about the precise features of complex social or biological processes.

1.1.2 Fundamentals of Probability Theory

We now introduce the formal definitions and notation used to represent the basic elements of random generative processes. There are three formal components that together fully describe a random generative process. The first is the *sample space*, denoted by Ω . The sample space is the set of all possible⁵ outcomes of the random generative process. Individual outcomes (sometimes known as *sample points*) are denoted by $\omega \in \Omega$. Outcomes can be represented by numbers, letters, words, or other symbols—whatever is most convenient for describing every distinct possible outcome of the random generative process. For example, if we wanted to describe a single roll of a six-sided die, we could let $\Omega = \{1, 2, 3, 4, 5, 6\}$. To describe a roll of two six-sided dice, we could let Ω be the set of all ordered pairs of integers between 1 and 6, that is, $\Omega = \{(x, y) \in \mathbb{Z}^2 : 1 \leq x \leq 6, 1 \leq y \leq 6\}$. To describe a fair coin flip, we could let $\Omega = \{\text{Heads}, \text{Tails}\}$ or $\Omega = \{H, T\}$. To describe choosing a random person in the United States and measuring their height in inches, we could let Ω be the set of all positive real numbers, $\Omega = \mathbb{R}^+$.

The second component of a random generative process is the *event space*. Events are subsets of Ω and are denoted by capital Roman letters, for example, $A \subseteq \Omega$. Whereas Ω describes all distinguishable states of the world that could result from the generative process, an event may occur in multiple states of the world, so we represent it as a set containing all states of the world in which it occurs. For example, in the case of rolling a single six-sided die, we could represent the event of rolling an even number by the set $A = \{\omega \in \Omega : \omega \text{ is even}\} = \{2, 4, 6\}$. Of course, an event can also correspond to a single state of the world, for instance, the event of rolling a 3, which we might represent by

³ See Diaconis, Holmes, and Montgomery (2007).

⁴ Current thinking in physics suggests that randomness *is* a fundamental feature of quantum-mechanical processes rather than merely a representation of unknown underlying variables determining individual outcomes. We are not considering quantum mechanics in this book. Suffice it to say that quantum randomness is probably not relevant to the social or health sciences.

⁵ We use the word “possible” here loosely, as the probability of a given outcome occurring may be zero.

the set $B = \{3\}$; such events are variously known as *atomic events*, *elementary events*, or *simple events*. For a given random generative process, a set of events is called an event space if it satisfies certain properties, which we state in the following definition. (We use the notation A^C to refer to the *complement* of the event A with respect to the sample space: $A^C = \Omega \setminus A = \{\omega \in \Omega : \omega \notin A\}$.)

Definition 1.1.1. Event Space

A set S of subsets of Ω is an *event space* if it satisfies the following:

- Nonempty: $S \neq \emptyset$.
- Closed under complements: if $A \in S$, then $A^C \in S$.
- Closed under countable unions: if $A_1, A_2, A_3, \dots \in S$, then $A_1 \cup A_2 \cup A_3 \cup \dots \in S$.⁶

Since each event in an event space will have an associated probability of occurring, these properties ensure that certain types of events will always have well-defined probabilities. Consider, again, a single roll of a six-sided die. Suppose A is the event of rolling an even number. If we can assign a probability to this event occurring, then we will also be able to assign a probability to this event *not* occurring, that is, A^C , the event of rolling an odd number. Similarly, suppose B is the event of rolling a number greater than 4. If we can assign probabilities to A and B , then we will also be able to assign a probability to the event of *at least one* of these occurring, that is, $A \cup B$, the event of rolling a 2, 4, 5, or 6.

This brings us to the final component needed to mathematically describe a random generative process: the *probability measure*. A probability measure is a function $P : S \rightarrow \mathbb{R}$ that assigns a probability to every event in the event space.⁷ To ensure that P assigns probabilities to events in a manner that is coherent and in accord with basic intuitions about probabilities, we must place some conditions on P . Such conditions are provided by the *Kolmogorov probability axioms*, which serve as the foundation of probability theory. These axioms define a *probability space*, a construct that both accords with basic intuitions about probabilities and lends itself to rigorous and useful mathematics.

⁶ A set S of subsets of another set Ω that satisfies these properties is formally known as a σ -algebra or σ -field on Ω . Some readers well versed in set theory may wonder why we do not simply let $S = \mathcal{P}(\Omega)$, the power set (that is, the set of all subsets) of Ω . For reasons that we will not discuss in this book, this does not always work; for some sample spaces Ω , it is impossible to define the probability of every subset of Ω in a manner consistent with the axioms of probability (see Definition 1.1.2). We need not worry too much about this point, though; in practice we will be able to define the probability of any event of interest without much difficulty. For this reason, we suggest that readers not worry too much about σ -algebras on their first read.

⁷ Note: we do not make the stronger assumption that $P : S \rightarrow [0, 1]$, since we prove this in Theorem 1.1.4.

Definition 1.1.2. *Kolmogorov Axioms*

Let Ω be a sample space, S be an event space, and P be a probability measure. Then (Ω, S, P) is a *probability space* if it satisfies the following:

- Non-negativity: $\forall A \in S, P(A) \geq 0$, where $P(A)$ is finite and real.
- Unitarity: $P(\Omega) = 1$.
- Countable additivity: if $A_1, A_2, A_3, \dots \in S$ are pairwise disjoint,⁸ then

$$P(A_1 \cup A_2 \cup A_3 \cup \dots) = P(A_1) + P(A_2) + P(A_3) + \dots = \sum_i P(A_i).$$

The intuition behind each of these axioms is as follows: The first axiom states that the probability of any event is a non-negative number; there cannot be a less-than-zero chance of an event occurring. The second axiom states that the probability measure of the entire sample space is one.⁹ In other words, it is certain that *some* outcome will occur. Finally, the third axiom states that, given any number of *mutually exclusive* events, the probability that one of those events will occur is the sum of their individual probabilities. Together, these axioms are sufficient to rule out any probability statements that would be nonsensical, and they provide the building blocks that will allow us to derive other useful properties (as we will see in Theorem 1.1.4, *Basic Properties of Probability*).

We can represent any random generative process as a probability space (Ω, S, P) , as illustrated by the following simple example.

Example 1.1.3. *A Fair Coin Flip*

Consider a fair coin flip. Let H represent the outcome “heads” and T represent the outcome “tails.” Let $\Omega = \{H, T\}$ and $S = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$. Then we can define the probability measure as follows:

$$P(A) = \frac{1}{2} |A|, \forall A \in S.$$

The notation $|A|$ denotes the *cardinality* of the set A , that is, the number of elements in A . So this means

- $P(\emptyset) = \frac{1}{2} |\emptyset| = \frac{1}{2} \cdot 0 = 0$. The probability of nothing happening is zero.
- $P(\{H\}) = \frac{1}{2} |\{H\}| = \frac{1}{2} \cdot 1 = \frac{1}{2}$. The probability of getting heads is $\frac{1}{2}$.
- $P(\{T\}) = \frac{1}{2} |\{T\}| = \frac{1}{2} \cdot 1 = \frac{1}{2}$. The probability of getting tails is $\frac{1}{2}$.

⁸ Recall that sets A and B are disjoint if $A \cap B = \emptyset$. We say that $A_1, A_2, A_3, \dots$ are pairwise disjoint if each of them is disjoint from every other, that is, $\forall i \neq j, A_i \cap A_j = \emptyset$.

⁹ Notice that Definition 1.1.1 implies that any event space must contain Ω : S is nonempty, so $\exists A \in S$. Since S is closed under complements, $A^C \in S$, and so since S is closed under countable unions, $A \cup A^C = \Omega \in S$. Likewise, $\Omega^C = \emptyset \in S$.

- $P(\{H, T\}) = \frac{1}{2} |\{H, T\}| = \frac{1}{2} \cdot 2 = 1$. The probability of getting either heads or tails is one.

The reader can verify that S is a proper event space (that is, it is nonempty and closed under complements and countable unions) and that P satisfies the Kolmogorov axioms, so (Ω, S, P) is a probability space. Δ^{10}

Several other fundamental properties of probability follow directly from the Kolmogorov axioms.

Theorem 1.1.4. *Basic Properties of Probability*

Let (Ω, S, P) be a probability space.¹¹ Then

- Monotonicity: $\forall A, B \in S$, if $A \subseteq B$, then $P(A) \leq P(B)$.
- Subtraction rule: $\forall A, B \in S$, if $A \subseteq B$, then $P(B \setminus A) = P(B) - P(A)$.
- Zero probability of the empty set: $P(\emptyset) = 0$.
- Probability bounds: $\forall A \in S$, $0 \leq P(A) \leq 1$.
- Complement rule: $\forall A \in S$, $P(A^C) = 1 - P(A)$.

Proof: Let $A, B \in S$ with $A \subseteq B$. Since $B = A \cup (B \setminus A)$, and A and $(B \setminus A)$ are disjoint, countable additivity implies

$$P(B) = P(A) + P(B \setminus A).$$

Rearranging this equation, non-negative probabilities then imply monotonicity: $P(B \setminus A) \geq 0$, so

$$P(A) = P(B) - P(B \setminus A) \leq P(B).$$

Rearranging again yields the subtraction rule:

$$P(B \setminus A) = P(B) - P(A).$$

The subtraction rule, in turn, implies zero probability of the empty set: $A \subseteq A$, so

$$P(\emptyset) = P(A \setminus A) = P(A) - P(A) = 0.$$

Monotonicity and unitarity (and non-negativity) imply the probability bounds: since $A \subseteq \Omega$,

$$0 \leq P(A) \leq P(\Omega) = 1.$$

Finally, the subtraction rule and unitarity imply the complement rule:

$$P(A^C) = P(\Omega \setminus A) = P(\Omega) - P(A) = 1 - P(A). \quad \square$$

¹⁰ Note that we use the Δ symbol to denote the end of an example.

¹¹ This assumption shall henceforth be implicit in all definitions and theorems referring to Ω , S , and/or P .

We can put each of these properties in simple terms. Monotonicity implies that, if one event is a subset of another (so that the former always occurs whenever the latter does), then the probability of the former occurring is no greater than that of the latter. The subtraction rule implies that the probability that the second event occurs but not the first is equal to the probability of the second event minus the probability of the first event. Zero probability of the empty set means that *some* event in our event space must occur, and probability bounds mean that each of these events has some probability of occurring between zero and one. Finally, the complement rule implies that the probability of any of these events *not* occurring is one minus the probability of the event occurring—so that the probability that a given event either occurs or does not occur is one.

1.1.3 Joint and Conditional Probabilities

We often want to describe how the probability of one event relates to the probability of another. We begin by establishing the *joint probability* of events A and B , or the probability that events A and B will both occur in a single draw from (Ω, S, P) .

Definition 1.1.5. *Joint Probability*

For $A, B \in S$, the *joint probability* of A and B is $P(A \cap B)$.

In other words, the joint probability of two events A and B is the probability of the intersection of A and B (which is itself an event in S), that is, the set of all states of the world in which both A and B occur. We illustrate this point with the following example.

Example 1.1.6. *A Fair Die Roll*

Consider a roll of one fair (six-sided) die. Let $\Omega = \{1, 2, 3, 4, 5, 6\}$, $S = \mathcal{P}(\Omega)$ (the power set—that is, the set of all subsets—of Ω), and $P(A) = \frac{1}{6}|A|$, $\forall A \in S$. Let $A = \{\omega \in \Omega : \omega \geq 4\} = \{4, 5, 6\}$ and $B = \{\omega \in \Omega : \omega \text{ is even}\} = \{2, 4, 6\}$. Then

$$P(A \cap B) = P(\{4, 5, 6\} \cap \{2, 4, 6\}) = P(\{4, 6\}) = \frac{1}{6}|\{4, 6\}| = \frac{2}{6} = \frac{1}{3}. \triangle$$

Just as $P(A \cap B)$ is the probability that both A and B will occur in a single draw from (Ω, S, P) , $P(A \cup B)$ is the probability that A or B (or both) will occur in a single draw from (Ω, S, P) . The following theorem allows us to relate these two probabilities.

Theorem 1.1.7. *Addition Rule*

For $A, B \in S$,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Proof: Note that $(A \setminus B)$, $(B \setminus A)$, and $(A \cap B)$ are pairwise disjoint and

$$(A \cup B) = (A \setminus B) \cup (B \setminus A) \cup (A \cap B),$$

so by countable additivity,

$$\begin{aligned} P(A \cup B) &= P(A \setminus B) + P(B \setminus A) + P(A \cap B) \\ &= P(A \setminus (A \cap B)) + P(B \setminus (A \cap B)) + P(A \cap B) \\ &= P(A) - P(A \cap B) + P(B) - P(A \cap B) + P(A \cap B) \\ &= P(A) + P(B) - P(A \cap B), \end{aligned}$$

where the second equality holds because

$$\begin{aligned} A \setminus B &= \emptyset \cup (A \setminus B) \\ &= (A \cap A^c) \cup (A \cap B^c) \\ &= A \cap (A^c \cup B^c) \\ &= A \cap (A \cap B)^c \\ &= A \setminus (A \cap B), \end{aligned}$$

and likewise $B \setminus A = B \setminus (A \cap B)$, while the third equality follows from the subtraction rule, since $A \cap B \subseteq A$ and $A \cap B \subseteq B$. $\square$

In other words, the probability of *at least one* of two events occurring is equal to the sum of the probabilities of *each* occurring minus the probability of *both* occurring. Of course, if A and B are disjoint, this reduces to $P(A \cup B) = P(A) + P(B)$, which is just a special case of countable additivity.

We might also want to describe the probability of observing event A *given* that we observe event B . This is known as *conditional probability*.

Definition 1.1.8. Conditional Probability

For $A, B \in \mathcal{S}$ with $P(B) > 0$, the *conditional probability* of A given B is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

We can rearrange this definition to obtain another useful formula: the *Multiplicative Law of Probability*.

Theorem 1.1.9. Multiplicative Law of Probability

For $A, B \in \mathcal{S}$ with $P(B) > 0$,

$$P(A|B)P(B) = P(A \cap B).$$

Proof: Rearrange Definition 1.1.8. $\square$

One of the most important theorems regarding conditional probability is *Bayes' Rule* (also known as *Bayes' Theorem* or *Bayes' Law*). Bayes' Rule relates the conditional probability of A given B to the conditional probability of B given A . Suppose we have a hypothesis about the probability that some event A will occur, and we then observe event B . We can update the probability that we predict A will occur, using information about the frequency with which B occurs given A . This kind of deduction constitutes a key element of probabilistic reasoning and thus has many applications in the social sciences.

Theorem 1.1.10. Bayes' Rule

For $A, B \in \mathcal{S}$ with $P(A) > 0$ and $P(B) > 0$,

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}.$$

Proof: By the Multiplicative Law of Probability, $P(A \cap B) = P(B|A)P(A)$. So, by the definition of conditional probability,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}. \quad \square$$

Although we will not be making use of Bayes' Rule in this book, we would be remiss to neglect it entirely. The following example illustrates the above concepts, including the utility of Bayes' Rule, by combining the previous examples of fair coins and fair dice.

Example 1.1.11. Flipping a Coin and Rolling a Die

Consider the following generative process. An experimenter flips a fair coin. If the coin comes up heads, the experimenter rolls a fair four-sided die. If the coin comes up tails, the experimenter rolls a fair six-sided die. The sample space can thus be represented by

$$\Omega = \{(H, 1), (H, 2), (H, 3), (H, 4), (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)\}.$$

Let A denote the event of observing heads, B denote the event of observing 3, and C denote the event of observing 6. Formally, $A = \{(H, 1), (H, 2), (H, 3), (H, 4)\}$, $B = \{(H, 3), (T, 3)\}$, and $C = \{(T, 6)\}$. What is the (joint) probability of observing heads and 3? The probability of observing heads is $P(A) = \frac{1}{2}$. Additionally, if heads is observed, then the experimenter rolls a fair four-sided die, so the probability of observing 3 *given that heads has been observed* is $P(B|A) = \frac{1}{4}$. So, by the Multiplicative Law of Probability,

$$P(A \cap B) = P(B|A)P(A) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}.$$

Likewise, the probability of observing tails and 3 is

$$P(A^c \cap B) = P(B|A^c)P(A^c) = \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{12}.$$

Since the experimenter can never roll a 6, if the coin comes up heads, the probability of observing heads and 6 is

$$P(A \cap C) = P(\emptyset) = 0.$$

The conditional probability of observing 3 given that heads (or tails) was observed is relatively straightforward, as we see above. But suppose we wanted to know the conditional probability that heads was observed given that 3 is observed. This is where Bayes' Rule is useful. We want to know $P(A|B)$. We know $P(B|A)$ and $P(A)$. What is $P(B)$? From countable additivity,

$$P(B) = P(A \cap B) + P(A^c \cap B) = \frac{1}{8} + \frac{1}{12} = \frac{5}{24}.$$

Thus, by Bayes' Rule,

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{\frac{1}{4} \cdot \frac{1}{2}}{\frac{5}{24}} = \frac{3}{5}. \triangle$$

The “trick” used above to calculate $P(B)$ is actually a special case of another important theorem, the *Law of Total Probability*. To state this theorem, we require the following definition.

Definition 1.1.12. *Partition*

If $A_1, A_2, A_3, \dots \in S$ are nonempty and pairwise disjoint, and $\Omega = A_1 \cup A_2 \cup A_3 \cup \dots$, then $\{A_1, A_2, A_3, \dots\}$ is a *partition* of Ω .

A partition divides the sample space into mutually exclusive and exhaustive categories or “bins.”¹² Every outcome in Ω is contained in exactly one A_i , so exactly one event A_i in the partition occurs for any draw from (Ω, S, P) .

Theorem 1.1.13. *Law of Total Probability*

If $\{A_1, A_2, A_3, \dots\}$ is a partition of Ω and $B \in S$, then

$$P(B) = \sum_i P(B \cap A_i).$$

If we also have $P(A_i) > 0$ for $i = 1, 2, 3, \dots$, then this can also be stated as

$$P(B) = \sum_i P(B|A_i)P(A_i).$$

Proof: If $\{A_1, A_2, A_3, \dots\}$ is a partition of Ω , then $\forall i \neq j$,

$$(B \cap A_i) \cap (B \cap A_j) = (B \cap B) \cap (A_i \cap A_j) = B \cap (A_i \cap A_j) = B \cap \emptyset = \emptyset.$$

¹² The number of bins may be finite or countably infinite.

So $(B \cap A_1), (B \cap A_2), (B \cap A_3), \dots$ are pairwise disjoint. Thus,

$$\begin{aligned} P(B) &= P(B \cap \Omega) \\ &= P(B \cap (A_1 \cup A_2 \cup A_3 \cup \dots)) \\ &= P((B \cap A_1) \cup (B \cap A_2) \cup (B \cap A_3) \cup \dots) \\ &= \sum_i P(B \cap A_i), \end{aligned}$$

where the last equality follows from countable additivity.

Additionally, if $P(A_i) > 0$ for $i = 1, 2, 3, \dots$, then we can apply the Multiplicative Law of Probability to each term in the summation to obtain

$$P(B) = \sum_i P(B|A_i)P(A_i). \quad \square$$

The probability of an event B is effectively a weighted average of the conditional probabilities of that event (B), where the weights are determined by the probabilities of the events that are being conditioned on $(A_1, A_2, A_3, \dots)$. Notice that for any event A , $\{A, A^C\}$ is a partition of Ω , so the method we used to calculate $P(B)$ above was indeed a special case of the Law of Total Probability. In fact, this “trick” is so often necessary for the application of Bayes’ Rule that it is frequently incorporated directly into the statement of the theorem.

Theorem 1.1.14. *Alternative Forms of Bayes’ Rule*

If $\{A_1, A_2, A_3, \dots\}$ is a partition of Ω with $P(A_i) > 0$ for $i = 1, 2, 3, \dots$, and $B \in \mathcal{S}$ with $P(B) > 0$, then

$$P(A_j|B) = \frac{P(B|A_j)P(A_j)}{\sum_i P(B \cap A_i)},$$

or equivalently,

$$P(A_j|B) = \frac{P(B|A_j)P(A_j)}{\sum_i P(B|A_i)P(A_i)}.$$

Proof: Apply each form of the Law of Total Probability to the denominator in Theorem 1.1.10 (*Bayes’ Rule*). $\square$

As we can see from Bayes’ Rule, sometimes knowing whether one event has occurred gives us more information about whether another event has occurred. When the events are *independent*, this is not the case.

1.1.4 Independence of Events

We now define *independence* of events under a random generative process (sometimes referred to as *statistical independence* or *stochastic independence*). Independence is important because it codifies the idea that events are unrelated. Independence will be central to ideas in Sections 1.3.4 (*Independence of Random Variables*), 2.2 (*Summary Features of Joint Distributions*), 2.2.5 (*CEFs and BLPs under Independence*), and 3.1 (*I.I.D. Random Variables*), as well as identification results in Chapters 6 and 7.

Definition 1.1.15. *Independence of Events*

Events $A, B \in \mathcal{S}$ are *independent* if $P(A \cap B) = P(A)P(B)$.

In Example 1.1.11 (*Flipping a Coin and Rolling a Die*), events A (observing heads) and B (observing 3) are *not* independent, since $P(A \cap B) = \frac{1}{8}$, but $P(A)P(B) = \frac{1}{2} \cdot \frac{5}{24} = \frac{5}{48}$. However, if we changed the example so that the experimenter rolled a six-sided die regardless of the outcome of the coin flip, then A and B would be independent.

The following useful theorem follows immediately from this definition and the definition of conditional probability.

Theorem 1.1.16. *Conditional Probability and Independence*

For $A, B \in \mathcal{S}$ with $P(B) > 0$, A and B are independent if and only if $P(A|B) = P(A)$.¹³

Proof: By Definition 1.1.15,

$$A \text{ and } B \text{ are independent} \iff P(A \cap B) = P(A)P(B).$$

By Theorem 1.1.9 (*Multiplicative Law of Probability*), $P(A \cap B) = P(A|B)P(B)$. Thus,

$$\begin{aligned} A \text{ and } B \text{ are independent} &\iff P(A|B)P(B) = P(A)P(B) \\ &\iff P(A|B) = P(A). \quad \square \end{aligned}$$

Loosely speaking, this theorem tells us that when A and B are independent, knowing whether or not B has occurred gives us *no information* about the probability that A has occurred. Referring again to Example 1.1.11 (*Flipping a Coin and Rolling a Die*), events A (observing heads) and B (observing 3) are *not* independent, since $P(A) = \frac{1}{2}$, but $P(A|B) = \frac{3}{5}$. The fact that B occurred tells

¹³ Some texts give this as the definition of independence and then prove as a theorem that A and B are independent if and only if $P(A \cap B) = P(A)P(B)$. This formulation is equivalent, and the proof is virtually identical.

us that it is more likely that A occurred, since, overall, we expect to observe A (heads) half of the time, but among instances in which B (a die roll of 3) is observed, we expect that A will have occurred in three fifths of those instances. More extremely, A (heads) and C (a die roll of 6) are not independent, since $P(A) = \frac{1}{2}$, but

$$P(A|C) = \frac{P(A \cap C)}{P(C)} = \frac{P(\emptyset)}{P(C)} = 0.$$

Overall, the probability of observing heads is one half, but if we know that a 6 was rolled, we can be *certain* that the coin came up tails.

Similarly, $P(B) = \frac{5}{24}$, but $P(B|A) = \frac{1}{4}$. Among all realizations of this random generative process, we expect to observe B (a die roll of 3) five twenty-fourths of the time, but among just those instances in which A (heads) is observed, we expect to observe B one quarter of the time. So, the fact that the coin came up heads increases the probability of observing a die roll of 3. Finally, by the Law of Total Probability,

$$P(C) = P(C \cap A) + P(C \cap A^c) = 0 + P(C|A^c)P(A^c) = \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{12}.$$

In other words, we can only observe C (a die roll of 6) when we observe A^c (not heads, that is, tails), so since tails will come up half the time, and then among those instances a 6 will be rolled one sixth of the time, overall we will observe C one twelfth of the time. But when we know A has occurred, C is not possible, so

$$P(C|A) = \frac{P(C \cap A)}{P(A)} = \frac{P(\emptyset)}{P(A)} = 0.$$

When the coin comes up heads, we know for sure that the experimenter will not roll a 6.

Again, if we were to change the example so that the experimenter rolled a six-sided die regardless of the outcome of the coin flip, then A and B would be independent. Knowing the outcome of the coin flip would then tell us nothing about the probability of observing any outcome of the die roll, and vice versa.

1.2 RANDOM VARIABLES

Many aspects of the world can be quantified. When a random generative process has outcomes that can be assigned real number values, its properties can more easily be described mathematically. A *random variable* is a variable that takes on a real value that is determined by a random generative process. In other words, random variables allow us to neatly represent random events by mapping those events to plain numbers. They are remarkable mathematical objects that will allow us to characterize probabilities in concise mathematical terms. We begin by providing a formal definition in the next section.

1.2.1 What Is a Random Variable?

Informally, we can think of a random variable as a container or placeholder for a quantity that has yet to be determined by a random generative process, just as ordinary algebraic variables represent unknown or unspecified quantities. However, the technical definition of a random variable is less intuitive and requires some careful discussion. Formally, a random variable is a real-valued function of the outcome of a random generative process.

Definition 1.2.1. *Random Variable*

A *random variable* is a function $X : \Omega \rightarrow \mathbb{R}$ such that, $\forall r \in \mathbb{R}$, $\{\omega \in \Omega : X(\omega) \leq r\} \in \mathcal{S}$.¹⁴

Recall that each $\omega \in \Omega$ denotes a state of the world, which may be represented by anything: numbers, letters, words, or other symbols, whatever notation is most convenient to describe all the distinct possible outcomes that could occur. A random variable maps each of these states of the world to a real number. Thus, it is often remarked that, technically, a random variable is neither random nor a variable, as it is merely a function. However, we can still informally think of a random variable as a variable that takes on a value determined by a random generative process; when the state of the world is ω , the random variable takes on the value $X(\omega)$. Indeed, the dependence on ω is typically suppressed, so X is understood to mean $X(\omega)$. Thus, for example, the event $\{X = 1\}$ should be understood to mean the set of states $\{\omega \in \Omega : X(\omega) = 1\}$.

There are two ways one can apply functions to random variables. The first is to use the value of $X(\omega)$ as input into another function g , with the result being another random variable. The second is to operate on the function $X(\cdot)$ itself, returning a value characterizing some feature of the function. We provide two formal definitions that we will delineate notationally.

Definition 1.2.2. *Function of a Random Variable*

Let $g : U \rightarrow \mathbb{R}$ be a function, where $X(\Omega) \subseteq U \subseteq \mathbb{R}$. Then, if $g \circ X : \Omega \rightarrow \mathbb{R}$ is a random variable, we say that g is a *function of X* , and write $g(X)$ to denote the random variable $g \circ X$.¹⁵

¹⁴ This regularity condition is necessary and sufficient to ensure that X is a measurable function. In particular, it ensures that we can define the cumulative distribution function of X (see Definition 1.2.11, *Cumulative Distribution Function*).

¹⁵ Though this definition might seem superfluous, it is necessary because not every function $g : U \rightarrow \mathbb{R}$ is sufficiently well behaved that $g(X)$ is a random variable. So when we refer to a function g of X , we implicitly mean that g satisfies the regularity conditions necessary for $g(X)$ to be a random variable. Also, note that $X(\Omega)$ is the *range* of the function X (see Glossary of Mathematical Notation).

Examples of functions of a random variable include $g(X) = X^2$ and

$$g(X) = \begin{cases} 1 & : X > 0 \\ 0 & : \text{otherwise.} \end{cases}$$

It is a rather general definition, and allows us to formally work with transformations of random variables as random variables in their own right.

Definition 1.2.3. Operator on a Random Variable

An operator A on a random variable maps the function $X(\cdot)$ to a real number, denoted by $A[X]$.¹⁶

We will use operators to summarize random variables; Chapter 2 is devoted to describing and defining certain types of operators that may be of scientific interest. Since $g(X)$ is itself a random variable, it is possible to define $A[g(X)]$ (see, e.g., Section 2.1.5, *Expectation of a Function of a Random Variable*). We will not formally generalize these definitions later when we consider random vectors (Section 1.3, *Bivariate Relationships*), but the logic carries through.

We can define events in S in terms of a random variable X . For example, we could let

- $A = \{\omega \in \Omega : X(\omega) = 1\}$.
- $B = \{\omega \in \Omega : X(\omega) \geq 0\}$.
- $C = \{\omega \in \Omega : X(\omega)^2 < 10, X(\omega) \neq 3\}$.

Or, stated in words, A is the event that X takes on the value of one, B is the event that X takes on a value greater than or equal to zero, and C is the event that X takes on a value such that X^2 is less than 10 *and* X is not 3.¹⁷ As shorthand, we typically drop the $\omega \in \Omega$ and denote events by their defining conditions on X . So, for the above examples, we would write

- $A = \{X = 1\}$.
- $B = \{X \geq 0\}$.
- $C = \{X^2 < 10, X \neq 3\}$.

Furthermore, we use $\Pr[\cdot]$ to denote the probability that some condition(s) on a random variable¹⁸ will hold, or, equivalently, the probability that the event defined by those conditions will occur. So, for the above examples,

¹⁶ Everything we say about random variables is implicitly in the context of a probability space (Ω, S, P) . A random variable $X(\cdot)$ is endowed with its probabilistic properties by the probability measure. Operators will summarize these properties, so they implicitly embed information from (Ω, S, P) .

¹⁷ When we list conditions separated by commas, the commas implicitly mean “and.”

¹⁸ Or variables; see Section 1.3 (*Bivariate Relationships*).

- $\Pr[X = 1] = P(A)$.
- $\Pr[X \geq 0] = P(B)$.
- $\Pr[X^2 < 10, X \neq 3] = P(C)$.

We use uppercase letters (for example, X, Y, Z, W) to denote random variables and lowercase letters (for example, x, y, z, w) to denote generic outcome¹⁹ values, in other words, as variables in the regular, algebraic sense. Thus, when we say $X = x$, this can be read as “the random variable X takes on the value x .”

1.2.2 Discrete Random Variables

It will be useful to delineate different types of random variables based on their mathematical properties. We will begin by providing a formal definition for one of the simplest types of random variables: the *discrete random variable*.

Definition 1.2.4. Discrete Random Variable

A random variable X is *discrete* if its range, $X(\Omega)$, is a countable set.

A discrete random variable is a random variable that only can take on a finite or countably infinite number of different values. As a result, the mathematics for discrete random variables tend to be relatively simple. However, although all real-world measurements are in practice discrete, when we have relatively fine-grained measurements, it becomes mathematically simpler to represent them as random variables that can take on a continuum of possible values and use calculus to describe their properties. In Section 1.2.4 (*Continuous Random Variables*), we will consider random variables of this kind, which have an uncountably large number of possible values.

Given a discrete random variable X , we can summarize the probability of each outcome x occurring with a *probability mass function (PMF)*.

Definition 1.2.5. Probability Mass Function (PMF)

For a discrete random variable X , the *probability mass function* of X is

$$f(x) = \Pr[X = x], \forall x \in \mathbb{R}.$$

We can apply the definition of the PMF to our familiar die roll example, showing the simplicity of the mathematical representation.

Example 1.2.6. A Fair Die Roll

Consider again a roll of one fair (six-sided) die. Let X take on the value of the

¹⁹ We will informally refer to elements $x \in X(\Omega)$ as outcomes, even though, strictly speaking, they are not necessarily elements of the sample space Ω , and, indeed, there may be more than one $\omega \in \Omega$ such that $X(\omega) = x$.

outcome of the die roll; that is, let $\Omega = \{1, 2, 3, 4, 5, 6\}$ and $X(\omega) = \omega$, $\forall \omega \in \Omega$. Then $\Pr[X = 1] = \Pr[X = 2] = \dots = \Pr[X = 6] = \frac{1}{6}$. Out of many die rolls, we expect each of the values 1 through 6 to come up one sixth of the time. Thus, the PMF of X is

$$f(x) = \begin{cases} \frac{1}{6} & : x \in \{1, 2, 3, 4, 5, 6\} \\ 0 & : \text{otherwise.} \end{cases} \quad \Delta$$

We can also highlight a generalization of our coin flip example: a biased coin flip. The resulting random variable is known as a *Bernoulli* random variable.

Example 1.2.7. *A Biased Coin Flip (Bernoulli Distribution)*

Consider a coin flip with a (potentially) biased coin—that is, a coin that comes up tails with some unknown probability. Let $X = 0$ if the coin comes up heads and $X = 1$ if the coin comes up tails; that is, let $\Omega = \{H, T\}$ and let $X(H) = 0$ and $X(T) = 1$. Let p be the probability that the coin comes up tails: $\Pr[X = 1] = p$. Then $\Pr[X = 0] = 1 - \Pr[X = 1] = 1 - p$. Out of many coin flips, we expect that the proportion of times the coin comes up tails will be p and the proportion of times the coin comes up heads will be $1 - p$. The random variable X thus has the PMF

$$f(x) = \begin{cases} 1 - p & : x = 0 \\ p & : x = 1 \\ 0 & : \text{otherwise.} \end{cases} \quad \Delta$$

One further example illustrates how a discrete random variable can have a countably infinite number of possible values.

Example 1.2.8. *Flipping a Biased Coin Until It Comes Up Tails (Geometric Distribution)* Suppose we flipped a (potentially) biased coin repeatedly until the first time it came up tails. Let p be the probability that the coin comes up tails, and assume $0 < p < 1$. Let X be the number of flips it takes to get tails. For any given positive integer x , getting the first tails on exactly the x^{th} flip requires getting heads on each of the first $x - 1$ flips and then tails on the x^{th} . The probability of this happening is $(1 - p)^{x-1}p$, or the product of the probabilities of the desired outcome on each flip (since the flips are independent). So the PMF of X is

$$f(x) = \begin{cases} (1 - p)^{x-1}p & : x \in \mathbb{N} \\ 0 & : \text{otherwise.} \end{cases}$$

So, for example, if it is a fair coin (that is, $p = 1 - p = \frac{1}{2}$) then $\forall x \in \mathbb{N}$, $\Pr[X = x] = (\frac{1}{2})^x$; the first tails will be obtained on the first flip with probability $\frac{1}{2}$, on the second flip with probability $\frac{1}{4}$, on the third flip with probability $\frac{1}{8}$, and so on. Thus, X can take on *any* positive integer value, albeit with vanishingly small probability for large values. Δ

For a discrete random variable X , the PMF tells us everything about its *distribution*, which is loosely defined as the collection of probabilities assigned to events that can be defined just in terms of X . To illustrate how the PMF can fully describe a discrete random variable, consider a random variable X such that $f(x) = 0$ for all $x \notin \mathbb{Z}$ (that is, X takes on only integer values). Then

- $\Pr[X \geq 3] = \sum_{x=3}^{\infty} f(x).$
- $\Pr[X \geq 3 \text{ or } X = 1] = f(1) + \sum_{x=3}^{\infty} f(x).$
- $\Pr[X < 4] = \sum_{x=1}^3 f(x).$

For example, for the die roll (Example 1.2.6),

- $\Pr[X \geq 3] = \sum_{x=3}^6 f(x) = \frac{4}{6} = \frac{2}{3}.$
- $\Pr[X \geq 3 \text{ or } X = 1] = f(1) + \sum_{x=3}^6 f(x) = \frac{1}{6} + \frac{4}{6} = \frac{5}{6}.$
- $\Pr[X < 4] = \sum_{x=1}^3 f(x) = \frac{3}{6} = \frac{1}{2}.$

We now note some important properties of PMFs that accord with our general intuitions about probabilities.

Theorem 1.2.9. *Properties of PMFs*

For a discrete random variable X with PMF f ,

- $\forall x \in \mathbb{R}, f(x) \geq 0.$
- $\sum_{x \in X(\Omega)} f(x) = 1.$

We omit the proof of this theorem, but the intuition follows directly from the Kolmogorov axioms (Definition 1.1.2): none of the probabilities are negative, and all probabilities add up to one. More generally, the following theorem gives the formula for using the PMF to compute the probability of *any* event defined in terms of a discrete random variable X .

Theorem 1.2.10. *Event Probabilities for Discrete Random Variables*

For a discrete random variable X with PMF f , if $D \subseteq \mathbb{R}$ and $A = \{X \in D\}$, then

$$P(A) = \Pr[X \in D] = \sum_{x \in X(A)} f(x).$$

We also omit the proof of this theorem, though again, it follows directly from the Kolmogorov axioms. Note that any condition on X can be expressed as $X \in D$ for some set $D \subseteq \mathbb{R}$, so this theorem allows us to compute the probability of any event defined in terms of a discrete random variable X . Theorem 1.2.10 thus establishes the link between our results on event probabilities from Section 1.1.2 (*Fundamentals of Probability Theory*) and random variables.

1.2.3 Cumulative Distribution Functions

In the previous section, we defined the concept of a PMF, which describes the distribution of a discrete random variable. We now define an alternative (and, as we will see, more general) way of describing the distribution of a random variable, the *cumulative distribution function* (CDF). All random variables (such as the *continuous* random variables we will consider in Section 1.2.4, *Continuous Random Variables*), not just discrete ones, have CDFs.

Definition 1.2.11. *Cumulative Distribution Function (CDF)*

For a random variable X , the *cumulative distribution function* of X is

$$F(x) = \Pr[X \leq x], \forall x \in \mathbb{R}.$$

The CDF returns the probability that an outcome for a random variable will be less than or equal to a given value. Importantly, given any random variable X , the CDF of X tells us *everything* there is to know about the behavior of X . For any event A that can be described just in terms of X , we can derive the probability of A from the CDF of X alone. (This, again, is what we mean when we talk about the distribution of a random variable X : the complete collection of probabilities assigned to events defined in terms of X .) The following important properties of CDFs follow immediately from the axioms and basic properties of probability.

Theorem 1.2.12. Properties of CDFs

For a random variable X with CDF F ,

- F is nondecreasing: $\forall x_1, x_2 \in \mathbb{R}$, if $x_1 < x_2$, then $F(x_1) \leq F(x_2)$.
- $\lim_{x \rightarrow -\infty} F(x) = 0$.
- $\lim_{x \rightarrow \infty} F(x) = 1$.
- $\forall x \in \mathbb{R}$, $1 - F(x) = \Pr[X > x]$.

Proof: Let $x_1, x_2 \in \mathbb{R}$ with $x_1 < x_2$. Then, since $\{X \leq x_1\} \subseteq \{X \leq x_2\}$, monotonicity implies that F is nondecreasing:

$$F(x_1) = \Pr[X \leq x_1] = P(\{X \leq x_1\}) \leq P(\{X \leq x_2\}) = \Pr[X \leq x_2] = F(x_2).$$

P is continuous,²⁰ and $\lim_{x \rightarrow -\infty} \{X \leq x\} = \{X \in \emptyset\} = \emptyset$, so

$$\begin{aligned} \lim_{x \rightarrow -\infty} F(x) &= \lim_{x \rightarrow -\infty} \Pr[X \leq x] \\ &= \lim_{x \rightarrow -\infty} P(\{X \leq x\}) \\ &= P\left(\lim_{x \rightarrow -\infty} \{X \leq x\}\right) \\ &= P(\emptyset) = 0. \end{aligned}$$

Similarly, $\lim_{x \rightarrow \infty} \{X \leq x\} = \{X \in \mathbb{R}\} = \Omega$, so

$$\lim_{x \rightarrow \infty} F(x) = \lim_{x \rightarrow \infty} \Pr[X \leq x] = \lim_{x \rightarrow \infty} P(\{X \leq x\}) = P\left(\lim_{x \rightarrow \infty} \{X \leq x\}\right) = P(\Omega) = 1.$$

Finally, by the complement rule, $\forall x \in \mathbb{R}$,

$$\begin{aligned} 1 - F(x) &= 1 - \Pr[X \leq x] \\ &= 1 - P(\{X \leq x\}) \\ &= P(\{X \leq x\}^c) \\ &= P(\{X > x\}) \\ &= \Pr[X > x]. \quad \square \end{aligned}$$

²⁰ More precisely, for any sequence of sets $A_1, A_2, A_3, \dots \in S$, if $A_1 \subseteq A_2 \subseteq A_3 \subseteq \dots$ and $A_1 \cup A_2 \cup A_3 \cup \dots = A$, then

$$\lim_{i \rightarrow \infty} P(A_1 \cup A_2 \cup \dots \cup A_i) = P(A),$$

and for any sequence of sets $B_1, B_2, B_3, \dots \in S$, if $B_1 \supseteq B_2 \supseteq B_3 \supseteq \dots$ and $B_1 \cap B_2 \cap B_3 \cap \dots = B$, then

$$\lim_{i \rightarrow \infty} P(B_1 \cap B_2 \cap \dots \cap B_i) = P(B).$$

This property allows us to “pass” the limit into $P(\cdot)$. We omit the proof of this fact.

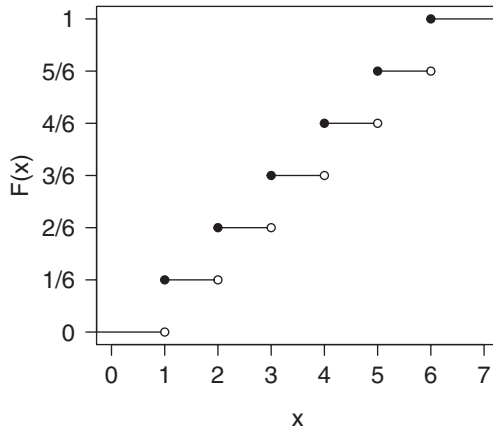


FIGURE 1.2.1. Die Roll CDF

Returning again to the die roll example, we can illustrate how the CDF characterizes the distribution of a random variable.

Example 1.2.13. CDF of a Fair Die Roll

For discrete random variables such as a die roll, the CDF is necessarily a step function, that is, a function that is flat everywhere except where it “jumps” from one value to another. Figure 1.2.1 shows the CDF for the die roll, which illustrates that the CDF increases by $\frac{1}{6}$ at each value in $\{1, 2, 3, 4, 5, 6\}$. We can evaluate the CDF at any real value, for example,

- $F(-1) = 0$.
- $F(1) = \frac{1}{6}$.
- $F(1.5) = \frac{1}{6}$.
- $F(2) = \frac{2}{6} = \frac{1}{3}$.
- $F(6) = 1$.
- $F(7) = 1$.
- $F(24,603) = 1$.

Note that, in this example, the value of the CDF for any x greater than or equal to 6 will be one; since any outcome we get is guaranteed to be less than or equal to 6, it follows that any outcome we get is guaranteed to be less than or equal to 7 or 24,603.

Additionally, as usual, we can use the CDF to compute event probabilities:

- $\Pr[X < 2] = \Pr[X \leq 1] = F(1) = \frac{1}{6}$,
- $\Pr[X \geq 3] = 1 - \Pr[X < 3] = 1 - \Pr[X \leq 2] = 1 - F(2) = \frac{4}{6} = \frac{2}{3}$,
- $\Pr[2 \leq X \leq 4] = \Pr[X \leq 4] - \Pr[X < 2] = F(4) - F(1) = \frac{4}{6} - \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$,

where the second line applies the complement rule and the third applies the subtraction rule (see Theorem 1.1.4, *Basic Properties of Probability*). Δ

However, there are other types of random variables that are more complex than discrete random variables. But regardless of type, every random variable has a CDF. We take up one important special case in the following section, where we consider random variables that have continuous CDFs.

1.2.4 Continuous Random Variables

Sometimes we are interested in quantities that can be measured to essentially any degree of precision, such as height, weight, or time. When a random variable represents the outcome of such a measurement, the number of possible values it could take on is limited only by the precision of our measuring instruments. For reasons of mathematical convenience, such a random variable can be treated as having a continuum of possible values, and is thus known as a *continuous random variable*.²¹

Loosely speaking, a random variable is continuous if its CDF is a continuous function.²² This implies that a continuous random variable X can take on an uncountably infinite number of different values; that is, $X(\Omega)$ is an uncountable set (and therefore, so is Ω). More specifically, $X(\Omega)$ must include some interval or union of intervals of the real line. Formally, a continuous random variable is defined as follows:

Definition 1.2.14. *Continuous Random Variable*

A random variable X is *continuous* if there exists a non-negative function $f: \mathbb{R} \rightarrow \mathbb{R}$ such that the CDF of X is

$$F(x) = \Pr[X \leq x] = \int_{-\infty}^x f(u) du, \forall x \in \mathbb{R}.$$

The function f is called the *probability density function (PDF)*. The Fundamental Theorem of Calculus implies that, for any continuous random variable, the PDF is unique and can be defined as follows:

²¹ There are also random variables with distributions that share features of both discrete and continuous random variables. In practice, we can treat any distribution as continuous and use generalized integration. Since this requires measure theory, we do not discuss such distributions explicitly in this book. Nevertheless, while we often state definitions and theorems in this book only in terms of discrete and/or continuous distributions, readers should bear in mind that these can almost always be generalized to all probability distributions using the appropriate measure theory.

²² Technically, this is a necessary but not sufficient condition for a random variable to be continuous. The sufficient condition is that the corresponding probability measure be absolutely continuous with respect to the Lebesgue measure. This is a technical measure-theoretic condition that need not concern us, as it is implied by Definition 1.2.14.

Definition 1.2.15. *Probability Density Function (PDF)*

For a continuous random variable X with CDF F , the *probability density function* of X is

$$f(x) = \left. \frac{dF(u)}{du} \right|_{u=x}, \forall x \in \mathbb{R}.$$

Conceptually, the PDF is the continuous analog to the PMF in that it describes how the CDF changes with x . The difference is that, whereas a PMF specifies the size of the “jump” in the CDF at a point x , a PDF gives the instantaneous slope (derivative) of the CDF at a point x . That is, for a very small number $\epsilon > 0$, if we moved from x to $x + \epsilon$, the CDF would change by approximately $\epsilon f(x)$.

We now establish some basic properties of PDFs, which are analogous to those of PMFs (Theorem 1.2.9).

Theorem 1.2.16. *Properties of PDFs*

For a continuous random variable X with PDF f ,

- $\forall x \in \mathbb{R}, f(x) \geq 0$.
- $\int_{-\infty}^{\infty} f(x) dx = 1$.

Proof: Let F be the CDF of X . F is nondecreasing, so, $\forall x \in \mathbb{R}$,

$$f(x) = \left. \frac{dF(u)}{du} \right|_{u=x} \geq 0,$$

and

$$\int_{-\infty}^{\infty} f(x) dx = \lim_{x \rightarrow \infty} \int_{-\infty}^x f(u) du = \lim_{x \rightarrow \infty} F(x) = 1. \quad \square$$

Using these properties, we can relate continuous random variables to events, much as we did in Theorem 1.2.10 (*Event Probabilities for Discrete Random Variables*). The following theorem establishes that we can express any event of the form $\{X \in I\}$, where I is an interval in $\mathbb{R}$, in terms of integrals of the PDF.

Theorem 1.2.17. *Event Probabilities for Continuous Random Variables*

For a continuous random variable X with PDF f ,

- $\forall x \in \mathbb{R}, \Pr[X = x] = 0$.
- $\forall x \in \mathbb{R}, \Pr[X < x] = \Pr[X \leq x] = F(x) = \int_{-\infty}^x f(u) du$.
- $\forall x \in \mathbb{R}, \Pr[X > x] = \Pr[X \geq x] = 1 - F(x) = \int_x^{\infty} f(u) du$.

- $\forall a, b \in \mathbb{R}$ with $a \leq b$,

$$\begin{aligned}\Pr[a < X < b] &= \Pr[a \leq X < b] = \Pr[a < X \leq b] \\ &= \Pr[a \leq X \leq b] = F(b) - F(a) = \int_a^b f(x)dx.\end{aligned}$$

Proof: Let F be the CDF of X . To establish that, $\forall x \in \mathbb{R}$, $\Pr[X = x] = 0$, we will proceed with a proof by contradiction. Let $x \in \mathbb{R}$, and suppose that $\Pr[X = x] = P(\{X = x\}) > 0$. Then, since P is continuous and $\lim_{u \rightarrow x^-} \{X \leq u\} = \{X < x\}$,

$$\begin{aligned}\lim_{u \rightarrow x^-} F(u) &= \lim_{u \rightarrow x^-} \Pr[X \leq u] \\ &= \lim_{u \rightarrow x^-} P(\{X \leq u\}) \\ &= P\left(\lim_{u \rightarrow x^-} \{X \leq u\}\right) \\ &= P(\{X < x\}) \\ &< P(\{X < x\}) + P(\{X = x\}) \\ &= P(\{X \leq x\}) \\ &= \Pr[X \leq x] \\ &= F(x).\end{aligned}$$

Thus, F has a discontinuity at x , contradicting the assumption that X is continuous. So $\Pr[X = x] \leq 0$, and thus, by non-negativity of probability, $\Pr[X = x] = 0$. Then, $\forall x \in \mathbb{R}$,

$$\begin{aligned}\Pr[X < x] &= P(\{X < x\}) \\ &= P(\{X < x\}) + P(\{X = x\}) \\ &= P(\{X \leq x\}) \\ &= \Pr[X \leq x] \\ &= F(x) \\ &= \int_{-\infty}^x f(u)du.\end{aligned}$$

Similarly, $\forall x \in \mathbb{R}$,

$$\begin{aligned}\Pr[X \geq x] &= P(\{X \geq x\}) \\ &= P(\{X > x\}) + P(\{X = x\})\end{aligned}$$

$$\begin{aligned}
&= P(\{X > x\}) \\
&= \Pr[X > x] \\
&= 1 - F(x) \\
&= 1 - \int_{-\infty}^x f(u) du \\
&= \int_{-\infty}^{\infty} f(u) du - \int_{-\infty}^x f(u) du \\
&= \int_x^{\infty} f(u) du.
\end{aligned}$$

Finally, by the same logic as above, $\forall a, b \in \mathbb{R}$ with $a < b$,

$$\Pr[a < X < b] = \Pr[a \leq X < b] = \Pr[a < X \leq b] = \Pr[a \leq X \leq b].$$

Furthermore, since

$$\{X < a\} \cup \{a \leq X \leq b\} = \{X \leq b\} \text{ and } \{X < a\} \cap \{a \leq X \leq b\} = \emptyset,$$

by countable additivity,

$$P(\{X < a\}) + P(\{a \leq X \leq b\}) = P(\{X \leq b\}).$$

Therefore,

$$P(\{a \leq X \leq b\}) = P(\{X \leq b\}) - P(\{X < a\}),$$

and thus

$$\begin{aligned}
\Pr[a \leq X \leq b] &= P(\{a \leq X \leq b\}) \\
&= P(\{X \leq b\}) - P(\{X < a\}) \\
&= \Pr[X \leq b] - \Pr[X < a] \\
&= F(b) - F(a) \\
&= \int_a^b f(x) dx,
\end{aligned}$$

where the final equality follows from the Fundamental Theorem of Calculus. $\square$

Note that by applying countable additivity, we can use Theorem 1.2.17 (*Event Probabilities for Continuous Random Variables*) to compute the probability of any event of the form $\{X \in D\}$, where D is a countable union of disjoint intervals in $\mathbb{R}$.²³

²³ In measure theory terms, for any Lebesgue-measurable set $D \subseteq \mathbb{R}$, $\Pr[X \in D] = \int_D f(x) dx$.

It may seem strange that, for a continuous random variable X , any specific outcome $x \in \mathbb{R}$ has probability $\Pr[X = x] = 0$. There is not necessarily a physical interpretation of this property; it is a consequence of the mathematical construct. In practice, our measurements are discrete because we do not have infinite precision of measurement. Nevertheless, continuous random variables can be thought of as mathematical approximations to random variables that are measured with sufficiently fine granularity that the gaps between possible outcome values are very small, and the probability of observing any one *exact* outcome is negligible.

Theorem 1.2.17 provides us with a clearer intuition for the meaning of the PDF: the area under the PDF over an interval is equal to the probability that the random variable takes on a value in that interval. Formally, this theorem implies that we need not worry about strict versus weak inequalities when describing the probability that a continuous random variable falls within a certain interval.

We now discuss two important examples of continuous distributions:²⁴ the *standard uniform distribution* and *standard normal distribution*.²⁵ We begin with the standard uniform distribution because of its simplicity.

Example 1.2.18. Standard Uniform Distribution

Consider the standard uniform distribution, denoted by $U(0, 1)$. Informally, if a random variable X follows the standard uniform distribution, X takes on a random real value from the interval $[0, 1]$, with all values in this interval equally likely to occur. We write $X \sim U(0, 1)$ to denote that X follows the standard uniform distribution. The standard uniform distribution has the PDF

$$f(x) = \begin{cases} 1 & : 0 \leq x \leq 1 \\ 0 & : \text{otherwise.} \end{cases}$$

Figure 1.2.2 plots the PDF of the standard uniform distribution.

The CDF of $U(0, 1)$ is thus

$$F(x) = \Pr[X \leq x] = \int_{-\infty}^x f(u) du = \begin{cases} 0 & : x < 0 \\ x & : 0 \leq x \leq 1 \\ 1 & : x > 1. \end{cases}$$

Figure 1.2.3 shows the CDF of the standard uniform distribution. $\triangle$

Example 1.2.19. Standard Normal Distribution

The reader may be familiar with the standard normal (or Gaussian) distribution. We will discuss some of the reasons why this distribution is important

²⁴ By *continuous distribution*, we simply mean the distribution of a continuous random variable.

²⁵ We will discuss other families of distributions in Chapter 5 (*Parametric Models*), but we do not need to establish the properties of, say, the Poisson or Gamma distribution to proceed. For a discussion of such distributions, see, e.g., Wackerly, Mendenhall, and Scheaffer (2008).

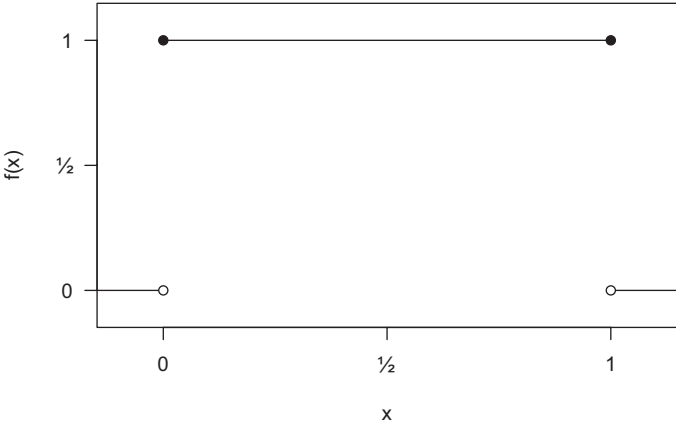


FIGURE 1.2.2. PDF of the Standard Uniform Distribution

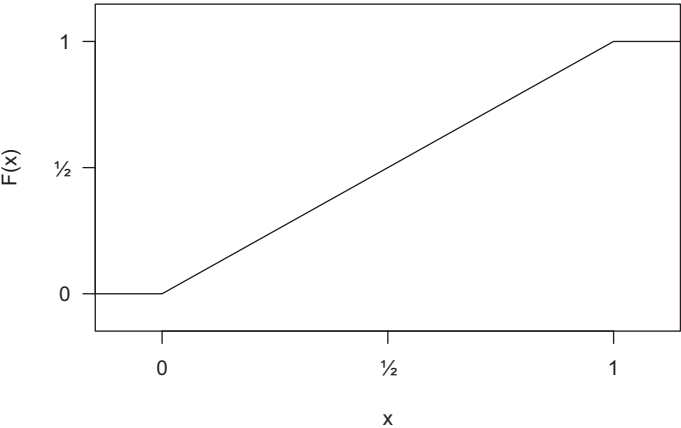


FIGURE 1.2.3. CDF of the Standard Uniform Distribution

to statistical inference later in this book (see, e.g., Section 3.2.4, *The Central Limit Theorem for Sample Means*). The standard normal distribution, denoted by $N(0, 1)$, has PDF

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}},$$

and CDF

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du.$$

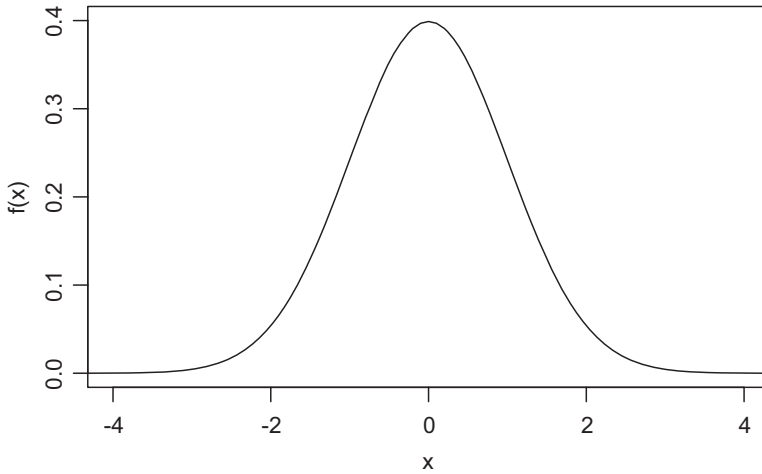


FIGURE 1.2.4. *PDF of the Standard Normal Distribution*

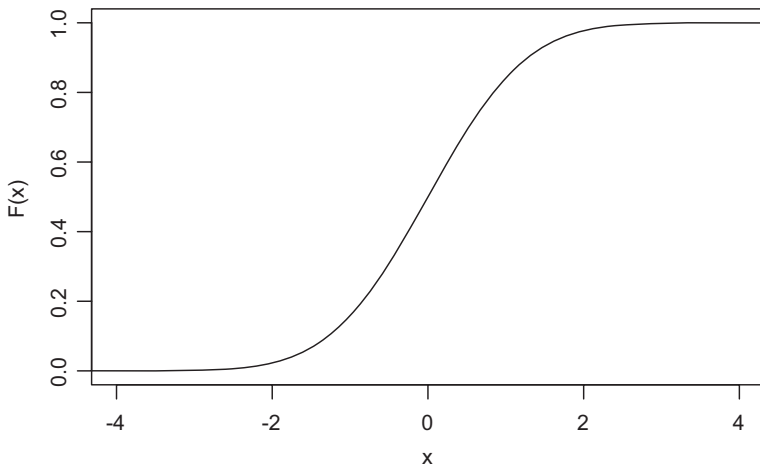


FIGURE 1.2.5. *CDF of the Standard Normal Distribution*

By convention, $\phi(x)$ and $\Phi(x)$ denote the PDF and CDF of the standard normal distribution, respectively. These are shown in Figures 1.2.4 and 1.2.5. $\triangle$

1.2.5 Support

We will often be interested in the subset of outcomes that could actually be observed for a given random variable. The set of values at which the PMF or PDF of a random variable is positive is called its *support*. In practice, we will

use this term to restrict the values under consideration from those on the real line to only those that could possibly be observed.

Definition 1.2.20. Support

For a random variable X with PMF/PDF f , the *support* of X is

$$\text{Supp}[X] = \{x \in \mathbb{R} : f(x) > 0\}.$$

For discrete X , $\text{Supp}[X]$ is the set of values that X takes on with nonzero probability. For example, if X is the outcome of a die roll, then $\text{Supp}[X] = \{1, 2, 3, 4, 5, 6\}$. For continuous X , $\text{Supp}[X]$ is the set of values over which X has nonzero probability density.²⁶ For example, if $X \sim U(0, 1)$, then $\text{Supp}[X] = [0, 1] = \{x \in \mathbb{R} : 0 \leq x \leq 1\}$. The notion of the support of a random variable will be particularly important in Section 2.2.3, when we define conditional expectations, and in Part III, in which assumptions about support will comprise key elements of our approach to identification.

1.3 BIVARIATE RELATIONSHIPS

As with random events, we often want to describe how multiple random variables are related. We therefore generalize the concepts presented above to the bivariate case. (Further generalization to the case of three or more random variables can be done analogously—see Section 1.4, *Multivariate Generalizations*.) We can have multiple random variables whose values are determined simultaneously by a single generative process: $X : \Omega \rightarrow \mathbb{R}$ and $Y : \Omega \rightarrow \mathbb{R}$, with X and Y each satisfying the regularity condition in Definition 1.2.1 (*Random Variable*).²⁷ Functions of and operators on multiple random variables are also defined analogously.

The following fundamental definition will be important going forward: when we say two random variables are equal, we mean that they are equal *as functions*; they assign the same value to every state of the world.

Definition 1.3.1. Equality of Random Variables

Let X and Y be random variables. Then $X = Y$ if, $\forall \omega \in \Omega$, $X(\omega) = Y(\omega)$.²⁸

²⁶ As Manski (1999, p. 15) intuitively describes it, “A value x_0 is said to be on the *support* of the probability distribution of x if there is a positive probability of observing x arbitrarily close to x_0 .”

²⁷ Whenever we have multiple random variables in a given setting, we always have a single probability space in the background.

²⁸ It is not technically equivalent, however, to say that $\Pr[X = Y] = 1$. Although $X = Y$ implies $\Pr[X = Y] = 1$, the converse is not strictly true in general, as there could be “measure zero” events for which $X \neq Y$. Consider, for example, a coin that always comes up heads. Let $\Omega = \{H, T\}$. Let $X(H) = 0$, $X(T) = 1$, $Y(H) = 0$, and $Y(T) = 2$. Then $\Pr[X = Y] = 1$, but $X \neq Y$. When $\Pr[X = Y] = 1$, X and Y are said to be *equal almost surely*.

From this definition, it follows that two functions of a random variable, $g(X)$ and $h(X)$, are equal as random variables if and only if they are equal as functions on $X(\Omega)$.

Theorem 1.3.2. *Equality of Functions of a Random Variable*

Let X be a random variable, and let f and g be functions of X . Then

$$g(X) = h(X) \iff \forall x \in X(\Omega), g(x) = h(x).$$

Proof: Suppose that $\forall x \in X(\Omega), g(x) = h(x)$. Let $\omega \in \Omega$. Then $X(\omega) \in X(\Omega)$, so $g(X(\omega)) = h(X(\omega))$. Thus, $\forall \omega \in \Omega, g(X(\omega)) = h(X(\omega))$, so $g(X) = h(X)$.

Now suppose that $g(X) = h(X)$, so $\forall \omega \in \Omega, g(X(\omega)) = h(X(\omega))$. Let $x \in X(\Omega)$. Then $\exists \omega \in \Omega$ such that $X(\omega) = x$, so $g(x) = h(x)$. Thus, $\forall x \in X(\Omega), g(x) = h(x)$. $\square$

This theorem will be especially important in Section 2.2.3, where it will allow us to make concise statements about conditional expectation functions.

1.3.1 Discrete Bivariate Distributions

Just as univariate random variables can be completely described by their PMF or PDF, bivariate distributions can be completely described by their joint PMF or PDF. To ground intuitions, we begin with the discrete case. In the case where X and Y are both discrete random variables, we can define the *joint PMF* of X and Y . The joint PMF at (x, y) is the probability that $X = x$ and $Y = y$ in a single realization of the generative process.

Definition 1.3.3. *Joint PMF*

For discrete random variables X and Y , the *joint PMF* of X and Y is

$$f(x, y) = \Pr[X = x, Y = y], \forall x, y \in \mathbb{R}.$$

Likewise, for any random variables X and Y , we can define their *joint CDF*. The joint CDF at (x, y) gives the probability of observing $X \leq x$ and $Y \leq y$ in a single realization of the generative process.

Definition 1.3.4. *Joint CDF*

For random variables X and Y , the *joint CDF* of X and Y is

$$F(x, y) = \Pr[X \leq x, Y \leq y], \forall x, y \in \mathbb{R}.$$

As with univariate distributions, *every* bivariate distribution, whether discrete or continuous, is characterized by a CDF. Furthermore, as in the univariate case, the joint CDF tells us everything about the behavior of X and Y .

Given any event A that can be described just in terms of X and Y , the probability of A can be derived from the joint CDF of X and Y . The following example returns to the coinflip and die roll to illustrate concepts in the discrete case.

Example 1.3.5. *Flipping a Coin and Rolling a Die*

Consider the generative process from Example 1.1.11, in which the experimenter flips a coin and then rolls either a four-sided or six-sided die depending on the outcome of the coin flip. Let $X = 0$ if the coin comes up heads and $X = 1$ if the coin comes up tails, and let Y be the value of the outcome of the die roll. Then the joint PMF of X and Y is

$$f(x, y) = \begin{cases} \frac{1}{8} & : x = 0, y \in \{1, 2, 3, 4\} \\ \frac{1}{12} & : x = 1, y \in \{1, 2, 3, 4, 5, 6\} \\ 0 & : \text{otherwise.} \end{cases}$$

For discrete random variables, the joint CDF is constructed by summing over the appropriate values of X and Y . For example,

$$\begin{aligned} F(1, 3) &= \sum_{x \leq 1} \sum_{y \leq 3} f(x, y) \\ &= f(0, 1) + f(0, 2) + f(0, 3) + f(1, 1) + f(1, 2) + f(1, 3) \\ &= \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} = \frac{5}{8}. \triangle \end{aligned}$$

1.3.2 Discrete Marginal and Conditional Distributions

In addition to the joint PMF and joint CDF, we can also describe the distribution of X and Y in terms of their marginal and conditional PMFs. The marginal distribution of a variable tells us only information about that variable, irrespective of its relationship with other variables. Accordingly, the *marginal PMF* of Y is simply the PMF of Y , ignoring the existence of X . The following theorem shows how a marginal PMF can be derived from a joint PMF.

Theorem 1.3.6. *Marginal PMF*

For discrete random variables X and Y with joint PMF f , the *marginal PMF* of Y is

$$f_Y(y) = \Pr[Y = y] = \sum_{x \in \text{Supp}[X]} f(x, y), \forall y \in \mathbb{R}.$$

(Henceforth we shall abbreviate $\sum_{x \in \text{Supp}[X]}$ as $\sum_x \cdot$.) We omit the proof, but it follows directly from the Law of Total Probability (Theorem 1.1.13): we

sum the joint probabilities of $X = x$ and $Y = y$ for every possible outcome for X to get the overall probability that $Y = y$. We can also compute the marginal PMF of X analogously.

The conditional distribution, by contrast, tells us the distribution of a given variable when we know another variable has a specific value. The *conditional PMF* of Y given X tells us the probability that a given value of Y will occur, *given that* a certain value of X occurs. That is, the conditional PMF of Y given $X = x$ tells us what the PMF of Y would be if we “threw out” all realizations of the generative process except those in which $X = x$.

Definition 1.3.7. Conditional PMF

For discrete random variables X and Y with joint PMF f , the *conditional PMF* of Y given $X = x$ is

$$f_{Y|X}(y|x) = \Pr[Y = y|X = x] = \frac{\Pr[X = x, Y = y]}{\Pr[X = x]} = \frac{f(x, y)}{f_X(x)},$$

$\forall y \in \mathbb{R}$ and $\forall x \in \text{Supp}[X]$.

Since $x \in \text{Supp}[X]$ is equivalent to $f_X(x) > 0$, this domain condition ensures that the denominator is nonzero. The conditional PMF of X given Y is defined analogously. Note that the definition of the conditional PMF is directly analogous to that of conditional probability. To illustrate the concept of the conditional PMF, we return to the example of flipping a coin and rolling a die.

Example 1.3.8. Flipping a Coin and Rolling a Die

In the coin flip and die roll example (Example 1.1.11), the marginal PMFs are

$$f_X(x) = \begin{cases} \frac{1}{2} & : x = 0 \\ \frac{1}{2} & : x = 1 \\ 0 & : \text{otherwise} \end{cases}$$

and

$$f_Y(y) = \begin{cases} \frac{5}{24} & : y \in \{1, 2, 3, 4\} \\ \frac{1}{12} & : y \in \{5, 6\} \\ 0 & : \text{otherwise,} \end{cases}$$

and the conditional PMFs are

$$f_{X|Y}(x|y) = \begin{cases} \frac{3}{5} & : x = 0, y \in \{1, 2, 3, 4\} \\ \frac{2}{5} & : x = 1, y \in \{1, 2, 3, 4\} \\ 1 & : x = 1, y \in \{5, 6\} \\ 0 & : \text{otherwise} \end{cases}$$

and

$$f_{Y|X}(y|x) = \begin{cases} \frac{1}{4} & : x = 0, y \in \{1, 2, 3, 4\} \\ \frac{1}{6} & : x = 1, y \in \{1, 2, 3, 4, 5, 6\} \\ 0 & : \text{otherwise.} \end{cases}$$

Note that we can operate on marginal PMFs in the same way that we would on univariate PMFs, since they *are* univariate PMFs. Similarly, conditional PMFs are univariate PMFs given any fixed value for the conditioning variable. So, in this example,

$$f_{X|Y}(x|3) = \begin{cases} \frac{3}{5} & : x = 0 \\ \frac{2}{5} & : x = 1 \\ 0 & : \text{otherwise} \end{cases}$$

is a univariate PMF. $\triangle$

Notice that we can rearrange Definition 1.3.7 (*Conditional PMF*) to obtain an analog to Theorem 1.1.9 (*Multiplicative Law of Probability*) for PMFs.

Theorem 1.3.9. *Multiplicative Law for PMFs*

Let X and Y be two discrete random variables with joint PMF f . Then, $\forall x \in \mathbb{R}$ and $\forall y \in \text{Supp}[Y]$,

$$f_{X|Y}(x|y)f_Y(y) = f(x, y).$$

Proof: Rearrange Definition 1.3.7. $\square$

1.3.3 Jointly Continuous Random Variables

We can extend the definitions of marginal and conditional distributions to the case where both random variables are continuous, and define marginal and conditional PDFs analogously. To do this, however, it is not technically sufficient to simply have X and Y be continuous. Instead, we require a slightly stronger condition: X and Y must be *jointly continuous*.

Definition 1.3.10. *Jointly Continuous Random Variables*

Two random variables X and Y are jointly continuous if there exists a non-negative function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ such that the joint CDF of X and Y is

$$F(x, y) = \Pr[X \leq x, Y \leq y] = \int_{-\infty}^x \int_{-\infty}^y f(u, v) dv du, \forall x, y \in \mathbb{R}.$$

The function f is called the *joint probability density function (joint PDF)*. Just as a single continuous random variable is characterized by a continuous CDF, two jointly continuous random variables are characterized by a continuous joint CDF. Additionally, just as taking the derivative of the CDF of a continuous random variable yields the PDF, taking the mixed second-order partial derivative of the joint CDF of two jointly continuous random variables yields the joint PDF.

Definition 1.3.11. *Joint PDF*

For jointly continuous random variables X and Y with joint CDF F , the *joint PDF* of X and Y is

$$f(x, y) = \left. \frac{\partial^2 F(u, v)}{\partial u \partial v} \right|_{u=x, v=y}, \forall x, y \in \mathbb{R}.$$

Without going into too much detail regarding partial derivatives, the joint PDF has roughly the same interpretation as the univariate PDF: if we perturb x or y a little bit, how much does the CDF change? Perhaps more intuitively, as with univariate continuous distributions, event probabilities are computed by integration: $\forall a, b, c, d \in \mathbb{R}$ with $a \leq b$ and $c \leq d$,

$$\Pr[a \leq X \leq b, c \leq Y \leq d] = \int_a^b \int_c^d f(x, y) dy dx.$$

That is, the volume under the PDF over a region equals the probability that X and Y take on values such that (X, Y) is a point in that region. Indeed, the probability of *any* event (not just those represented by rectangular regions) can be computed by integration.

Theorem 1.3.12. *Event Probabilities for Bivariate Continuous Distributions*

For jointly continuous random variables X and Y with joint PDF f , if $D \subseteq \mathbb{R}^2$, then

$$\Pr[(X, Y) \in D] = \iint_D f(x, y) dy dx.^{29}$$

We omit the proof of this theorem. So, for example, the probability that (X, Y) will fall within the triangular region $\{0 \leq X \leq 1, Y \leq X\}$ is

$$\Pr[0 \leq X \leq 1, Y \leq X] = \int_0^1 \int_0^x f(x, y) dy dx.$$

Consequently, properties analogous to those in Theorem 1.2.16 (*Properties of PDFs*) and Theorem 1.2.17 (*Event Probabilities for Continuous Random Variables*) apply to bivariate continuous distributions: the joint PDF is non-negative everywhere and integrates to one; exact outcomes have probability zero,³⁰ and therefore strict versus weak inequalities in event specifications make no difference; for example, $\Pr[X \geq 7, 3 \leq Y < 6] = \Pr[X > 7, 3 < Y \leq 6]$ if X and Y are jointly continuous.

We can now define marginal and conditional PDFs analogously to marginal and conditional PMFs. As in the discrete case, the *marginal PDF* of Y is just the PDF of Y , ignoring the existence of X . The following theorem shows how a marginal PDF can be derived from a joint PDF.

Theorem 1.3.13. *Marginal PDF*

For jointly continuous random variables X and Y with joint PDF f , the *marginal PDF* of Y is

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx, \forall y \in \mathbb{R}.$$

We omit the proof, but the intuition is analogous to the discrete case: we use integration to “sum” the joint density of X and Y over all values of x to get the overall density for Y at a given y . As before, the marginal PDF of X is computed analogously.

By contrast, the *conditional PDF* of X given Y is the PDF of X *given that* a certain value of Y occurs.

²⁹ As in the univariate case, D must be Lebesgue-measurable, or the integral does not exist.

³⁰ For bivariate continuous distributions, this means not only that $\Pr[X = x, Y = y] = 0$, but also that any event that specifies *either* $X = x$ *or* $Y = y$ has probability zero, for example, $\Pr[X > x, Y = y] = 0$. Indeed, for any $D \subseteq \mathbb{R}^2$ that has zero area (that is, Lebesgue measure zero), $\Pr[(X, Y) \in D] = 0$, for example, $\Pr[X = Y] = 0$, since the set $\{(x, y) \in \mathbb{R}^2 : x = y\}$ is a line.

Definition 1.3.14. Conditional PDF

For jointly continuous random variables X and Y with joint PDF f , the *conditional PDF* of Y given $X = x$ is

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)}, \forall y \in \mathbb{R} \text{ and } \forall x \in \text{Supp}[X].$$

Again, the domain condition $x \in \text{Supp}[X]$ ensures that the denominator is nonzero. The conditional PDF of X given $Y = y$ is again defined analogously. As in the discrete case, marginal PDFs are univariate PDFs, and conditional PDFs are univariate PDFs given any fixed value for the conditioning variable. Theorem 1.2.16 (*Properties of PDFs*) thus applies to all marginal PDFs and to all conditional PDFs given any fixed value of the conditioning variable.

Just as with PMFs (see Theorem 1.3.9, *Multiplicative Law for PMFs*), we can rearrange Definition 1.3.14 to obtain an analog to Theorem 1.1.9 (*Multiplicative Law of Probability*) for PDFs.

Theorem 1.3.15. Multiplicative Law for PDFs

Let X and Y be two jointly continuous random variables with joint PDF f . Then, $\forall x \in \mathbb{R}$ and $\forall y \in \text{Supp}[Y]$,

$$f_{X|Y}(x|y)f_Y(y) = f(x,y).$$

Proof: Rearrange Definition 1.3.14. $\square$

1.3.4 Independence of Random Variables

Like random events, random variables can be statistically independent, since random variables are just real-valued functions of outcomes. The intuitions are analogous. We conclude this section by defining independence of random variables and noting some important implications of independence.

Definition 1.3.16. Independence of Random Variables

Let X and Y be either two discrete random variables with joint PMF f or two jointly continuous random variables with joint PDF f . Then X and Y are *independent* if, $\forall x, y \in \mathbb{R}$,

$$f(x,y) = f_X(x)f_Y(y).$$

We write $X \perp\!\!\!\perp Y$ to denote that X and Y are independent.

Note the similarity to the definition of independence of events (Definition 1.1.15). Returning again to the coin flip and die roll example, we see

that X (the outcome of the coin flip, where $X = 0$ for heads and $X = 1$ for tails) and Y (the outcome of the die roll) clearly are *not* independent, since, as we have already shown, $f(0, 3) = \frac{1}{8} \neq \frac{1}{2} \cdot \frac{5}{24} = f_X(0)f_Y(3)$.

The following theorem states some properties that hold if and only if X and Y are independent.

Theorem 1.3.17. Implications of Independence (Part I)

Let X and Y be either two discrete random variables with joint PMF f or two jointly continuous random variables with joint PDF f . Then the following statements are equivalent (that is, each one implies all the others):

- $X \perp\!\!\!\perp Y$.
- $\forall x, y \in \mathbb{R}, f(x, y) = f_X(x)f_Y(y)$.
- $\forall x \in \mathbb{R}$ and $\forall y \in \text{Supp}[Y], f_{X|Y}(x|y) = f_X(x)$.
- $\forall D, E \subseteq \mathbb{R}$, the events $\{X \in D\}$ and $\{Y \in E\}$ are independent.
- For all functions g of X and h of Y , $g(X) \perp\!\!\!\perp h(Y)$.

We omit the proof of this theorem. Notice that the third statement is analogous to Theorem 1.1.16 (*Conditional Probability and Independence*); if X and Y are independent, knowing the outcome for Y gives us *no information* about the probability of any outcome for X , as the conditional distribution of X remains the same regardless of the value that Y takes on.

1.4 MULTIVARIATE GENERALIZATIONS

We conclude this chapter by briefly outlining the generalizations of some of the above concepts to the case where we have more than two random variables. We provide no examples or proofs in this section, as our goal is simply to illustrate that the concepts we have presented in detail in the bivariate case are easily extended. Essentially all omitted definitions, properties, and theorems can be generalized analogously.

We begin by defining a natural way to characterize a collection of multiple random variables: a *random vector* is a vector whose components are random variables. Beginning in Chapter 2, we will use this notation frequently when we are considering the relationships between multiple random variables.

Definition 1.4.1. Random Vector

A *random vector* (of length K) is a function $\mathbf{X}: \Omega \rightarrow \mathbb{R}^K$ such that, $\forall \omega \in \Omega$,

$$\mathbf{X}(\omega) = (X_{[1]}(\omega), X_{[2]}(\omega), \dots, X_{[K]}(\omega)),$$

where $\forall k \in \{1, 2, \dots, K\}$, $X_{[k]}$ is a random variable.

Here we must pause to provide some remarks on notation. All vectors (random or otherwise) in this book shall be denoted by boldface letters, and all vectors are row vectors unless otherwise indicated. Importantly, we use bracketed subscripts to denote distinct random variables or components of a random vector, as we will use plain subscripts to denote multiple, independent realizations of a single random variable beginning in Chapter 3. As usual, we will suppress the ω notation, so we will write, for instance, $\mathbf{X} = (X_{[1]}, X_{[2]}, \dots, X_{[K]})$. In some cases, components of a random vector may be represented by distinct letters for notational convenience, for example, (X, Y) or $(X_{[1]}, X_{[2]}, \dots, X_{[K]}, Y)$, the latter of which may also be denoted by $(\mathbf{X}, Y)$.

As in the bivariate case, the joint CDF of $X_{[1]}, X_{[2]}, \dots, X_{[K]}$ tells us everything about the behavior of these random variables. Importantly, it makes no difference whether we think of $X_{[1]}, X_{[2]}, \dots, X_{[K]}$ as separate random variables or as components of a random vector $\mathbf{X}$. They still have a joint distribution characterized by a joint CDF, which thus also fully characterizes the distribution of $\mathbf{X}$.

Definition 1.4.2. *Joint CDF (Multivariate Case)*

For random variables $X_{[1]}, X_{[2]}, \dots, X_{[K]}$, the *joint CDF* of $X_{[1]}, X_{[2]}, \dots, X_{[K]}$ is

$$F(x_{[1]}, x_{[2]}, \dots, x_{[K]}) = \Pr[X_{[1]} \leq x_{[1]}, X_{[2]} \leq x_{[2]}, \dots, X_{[K]} \leq x_{[K]}],$$

$$\forall x_{[1]}, x_{[2]}, \dots, x_{[K]} \in \mathbb{R}.$$

As before, the joint CDF at $(x_{[1]}, x_{[2]}, \dots, x_{[K]})$ gives the probability of observing $X_{[1]} \leq x_{[1]}, X_{[2]} \leq x_{[2]}, \dots, X_{[K]} \leq x_{[K]}$, in a single realization of the generative process. This can equivalently be written as $F(\mathbf{x}) = \Pr[\mathbf{X} \leq \mathbf{x}]$, $\forall \mathbf{x} \in \mathbb{R}^K$.

In the case where $X_{[1]}, X_{[2]}, \dots, X_{[K]}$ are all discrete random variables, we can define the joint PMF of $X_{[1]}, X_{[2]}, \dots, X_{[K]}$ just as before.

Definition 1.4.3. *Joint PMF (Multivariate Case)*

For discrete random variables $X_{[1]}, X_{[2]}, \dots, X_{[K]}$, the *joint PMF* of $X_{[1]}, X_{[2]}, \dots, X_{[K]}$ is

$$f(x_{[1]}, x_{[2]}, \dots, x_{[K]}) = \Pr[X_{[1]} = x_{[1]}, X_{[2]} = x_{[2]}, \dots, X_{[K]} = x_{[K]}],$$

$$\forall x_{[1]}, x_{[2]}, \dots, x_{[K]} \in \mathbb{R}.$$

Again, as before, the joint PMF at $(x_{[1]}, x_{[2]}, \dots, x_{[K]})$ is the probability that $X_{[1]} = x_{[1]}, X_{[2]} = x_{[2]}, \dots, X_{[K]} = x_{[K]}$ in a single realization of the generative process. This can equivalently be written as $f(\mathbf{x}) = \Pr[\mathbf{X} = \mathbf{x}]$, $\forall \mathbf{x} \in \mathbb{R}^K$. In the case where $X_{[1]}, X_{[2]}, \dots, X_{[K]}$ are all continuous random variables, we must first define joint continuity before we can define the joint PDF.

Definition 1.4.4. *Jointly Continuous Random Variables (Multivariate Case)*

The random variables $X_{[1]}, X_{[2]}, \dots, X_{[K]}$, are *jointly continuous* if there exists a non-negative function $f: \mathbb{R}^K \rightarrow \mathbb{R}$ such that the joint CDF F of $X_{[1]}, X_{[2]}, \dots, X_{[K]}$ is

$$F(x_{[1]}, x_{[2]}, \dots, x_{[K]}) = \int_{-\infty}^{x_{[1]}} \int_{-\infty}^{x_{[2]}} \cdots \int_{-\infty}^{x_{[K]}} f(u_{[1]}, u_{[2]}, \dots, u_{[K]}) du_{[K]} \cdots du_{[2]} du_{[1]},$$

$$\forall x_{[1]}, x_{[2]}, \dots, x_{[K]} \in \mathbb{R}.$$

The function f is the joint PDF. As in the bivariate case, we define the joint PDF more explicitly as follows:

Definition 1.4.5. *Joint PDF (Multivariate Case)*

For jointly continuous random variables $X_{[1]}, X_{[2]}, \dots, X_{[K]}$ with joint CDF F , the *joint PDF* of $X_{[1]}, X_{[2]}, \dots, X_{[K]}$ is

$$f(x_{[1]}, x_{[2]}, \dots, x_{[K]}) = \left. \frac{\partial^K F(u_{[1]}, u_{[2]}, \dots, u_{[K]})}{\partial u_{[1]} \partial u_{[2]} \cdots \partial u_{[K]}} \right|_{u_{[1]}=x_{[1]}, u_{[2]}=x_{[2]}, \dots, u_{[K]}=x_{[K]}},$$

$$\forall x_{[1]}, x_{[2]}, \dots, x_{[K]} \in \mathbb{R}.$$

The intuition is the same as in the bivariate case, and event probabilities are again obtained by integrating the PDF over the region that corresponds to that event.

We can also define marginal and conditional PMFs/PDFs in the multivariate case. Here it will be useful to distinguish one of the random variables from the others, so we state the following definitions for random variables denoted by $X_{[1]}, X_{[2]}, \dots, X_{[K]}$, and Y —which, again, could also be combined into the random vector $(\mathbf{X}, Y)$. The notational distinction between the X s and Y will also be important in later chapters for distinguishing between random variables that are used to predict (the X s) and the random variable that we seek to predict (Y).

Definition 1.4.6. *Marginal PMF/PDF (Multivariate Case)*

For discrete random variables $X_{[1]}, X_{[2]}, \dots, X_{[K]}$, and Y with joint PMF f , the *marginal PMF* of Y is

$$f_Y(y) = \Pr[Y = y] = \sum_{x_{[1]}} \sum_{x_{[2]}} \cdots \sum_{x_{[K]}} f(x_{[1]}, x_{[2]}, \dots, x_{[K]}, y), \forall y \in \mathbb{R}.$$

For jointly continuous random variables $X_{[1]}, X_{[2]}, \dots, X_{[K]}$, and Y with joint PDF f , the *marginal PDF* of Y is

$$f_Y(y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_{[1]}, x_{[2]}, \dots, x_{[K]}, y) dx_{[1]} dx_{[2]} \cdots dx_{[K]}, \forall y \in \mathbb{R}.$$

The idea behind the marginal PMF/PDF is as before: we sum (or integrate) over all possible outcomes for the X variables to obtain the univariate PMF/PDF of Y , that is, the PMF/PDF of Y ignoring the existence of the X s. This is actually just one of many types of marginal distributions we can obtain in the multivariate case. We could similarly sum (or integrate) over just Y to obtain the marginal (joint) distribution of the X s, ignoring the existence of Y . Let $f_X(\mathbf{x})$ denote the marginal (joint) distribution of $\mathbf{X}$. Then, in the discrete case,

$$f_X(\mathbf{x}) = \sum_y f(\mathbf{x}, y),$$

and in the continuous case,

$$f_X(\mathbf{x}) = \int_{-\infty}^{\infty} f(\mathbf{x}, y) dy,$$

where $f(\mathbf{x}, y)$ is just another way of denoting the joint PMF/PDF. Indeed, we could choose any arbitrary $k \leq K$ of these random variables to “marginalize out,” thereby obtaining the marginal (joint) distribution of the remaining variables.

As in the bivariate case, whereas a marginal PMF/PDF describes the distribution of a random variable (or vector) ignoring the existence of other random variables, a conditional PMF/PDF describes the distribution of a random variable (or vector) given that other random variables take on certain values.

Definition 1.4.7. Conditional PMF/PDF (Multivariate Case)

For discrete random variables $X_{[1]}, X_{[2]}, \dots, X_{[K]}$, and Y with joint PMF f , the *conditional PMF* of Y given $\mathbf{X} = \mathbf{x}$ is

$$f_{Y|\mathbf{X}}(y|\mathbf{x}) = \Pr[Y = y | \mathbf{X} = \mathbf{x}] = \frac{\Pr[Y = y, \mathbf{X} = \mathbf{x}]}{\Pr[\mathbf{X} = \mathbf{x}]} = \frac{f(\mathbf{x}, y)}{f_X(\mathbf{x})},$$

$\forall y \in \mathbb{R}$ and $\forall \mathbf{x} \in \text{Supp}[\mathbf{X}]$.

For jointly continuous random variables $X_{[1]}, X_{[2]}, \dots, X_{[K]}$, and Y with joint PDF f , the *conditional PDF* of Y given $\mathbf{X} = \mathbf{x}$ is

$$f_{Y|\mathbf{X}}(y|\mathbf{x}) = \frac{f(\mathbf{x}, y)}{f_X(\mathbf{x})}, \forall y \in \mathbb{R} \text{ and } \forall \mathbf{x} \in \text{Supp}[\mathbf{X}].$$

Similar to marginal distributions, we can in fact condition on any $k \leq K$ of the variables and obtain the conditional distribution of the others, for example, $f_{X|Y}$, the conditional (joint) distribution of X given Y .

1.5 FURTHER READINGS

Alternative readings covering much of our Chapter 1 are available at varying levels of technicality. We recommend Blitzstein and Hwang (2014) and Freedman, Pisani, and Purves (1998), both of which provide rigorous, but reader-friendly, introductions to probability theory. Freedman, Pisani, and Purves (1998) is especially recommended for readers without prior exposure to undergraduate-level probability or mathematics. Freedman and Stark (2003) is a highly accessible article that discusses the fundamentals and interpretation of probabilistic statements (with application to earthquake forecasts). For more traditional and in-depth introductions to mathematical statistics and probability, we recommend Rice (2007), Casella and Berger (2001), Wackerly, Mendenhall, and Scheaffer (2008), and Pitman (1993). Wasserman (2004) provides a clear and concise treatment of the materials covered here. Billingsley (1995) is a measure-theoretic treatment of probability theory that comes highly recommended for readers with previous exposure to real analysis at the advanced undergraduate level or above.